

Use-Case Flow Specification

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Use Case: Search & Reserve Equipment

Problem Statement #52 — Community Tool & Equipment Library

Actors: Community Member (primary), Library Custodian

Preconditions

1. The Community Member has a registered, active account with no overdue items on their record.
2. The system's tool inventory data reflects real-time availability across all depot lockers.

Postconditions

1. Success: A reservation record is created, the required security deposit is held, and the tool's status changes to "Reserved" for the selected pickup depot and time slot.
2. Failure: No reservation is created and the tool's availability status remains unchanged.

Main Success Scenario

1. Community Member logs in and searches the tool inventory by category, keyword, or depot location.
2. System displays matching tools with real-time availability across neighborhood depot lockers.
3. Community Member selects a desired tool and a pickup depot/time slot.
4. System checks the Member's account status (no overdue tools, no holds).
5. System includes the Calculate Deposit use case: it determines the required security deposit based on the tool category and enforces the maximum loan duration limit.
6. System presents the deposit amount, loan duration, and total cost summary to the Member.
7. Community Member confirms the reservation.
8. System places a deposit hold, marks the tool as "Reserved," and generates a confirmation with pickup instructions.
9. System sends a confirmation notification (email/SMS) to the Member. Use case ends successfully.

Alternate Flow

4a. Member Has Overdue Tools or Holds

- 4a1. System detects that the Member currently has overdue tools or an active hold on their account.
- 4a2. System rejects the reservation attempt and displays a message stating the reason (e.g., "Return overdue item before reserving new equipment").
- 4a3. System suggests the Member return the overdue tool or contact the Library Custodian to resolve the hold.
- 4a4. Use case ends without a reservation being created.